

Long-term development of transition hardwood and *Pinus strobus* - *Quercus* mixedwood forests with implications for future adaptation and mitigation potential

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ABSTRACT

Uncertainty about global environmental change has led to increased emphasis on the climate adaptation and mitigation potential of forests. Given the linkages between adaptive capacity and ecosystem complexity, this increased emphasis has motivated evaluations of the compositional, functional, and structural conditions characterizing a given forest ecosystem in the context of future stressors; however, less is known about how these conditions develop over time or vary between secondary forests shaped by a history of land use. To address this need, we capitalize on a 69-year field experiment to examine the long-term structural and compositional dynamics of transition ecotone hardwood (forests containing species from both southern and northern communities) and *Pinus strobus*-*Quercus* mixedwood (stands characterized by hardwood and softwood species mixtures) forests, communities which are commonly found in the Northeastern US. As expected, we observed a general increase in biomass over time and increasing density and structural complexity, with live aboveground biomass greatest in structurally complex mixedwood systems dominated by large diameter *P. strobus*. While all forests trended toward greater shade tolerance with increasing stand age, the functional identities of hardwood stands were stable over time compared to mixedwoods, which were more transient and ultimately generated trait profiles resembling hardwood-dominated stands. Ingrowth on all sites favored shade-tolerant *Fagus grandifolia* and *Tsuga canadensis*, which have lower future climate compatibility and adaptability compared to overstory trees, with a noticeable absence of *P. strobus* and *Quercus* spp. regeneration. Although all forest types exhibited some conditions that foster adaptation potential, the compositional stability of hardwood-dominated systems highlight the capacity of these stands to maintain comparable levels of adaptive capacity into the future. Conversely, given that *P. strobus*-*Quercus* mixedwoods examined are largely an artifact of land use, the natural successional patterns of these forests may lead to a reduction in these mixed species communities and a depreciation of associated climate benefits without silvicultural intervention to favor recruitment of these constituent species.

1. Introduction

Over the next century, temperate forests are expected to experience significant changes in structural and compositional conditions due to global change. As a result, numerous biophysical drivers may challenge the ability of forested systems to withstand and adapt to new conditions, such as changes in climate regimes (IPCC, 2014), introductions of invasive pests and pathogens (Lovett et al., 2016), and alterations in historic disturbance regimes (Dale et al., 2001). Consequently, interest in understanding forest structural and compositional development in the context of climate adaptation and mitigation has increased, with

particular focus on management strategies that create conditions that balance achievement of multiple objectives, such as commodity production, maintenance of critical ecosystem services (e.g., carbon sequestration), and increased adaptation potential to forecasted change (Joyce et al., 2009; Park et al., 2014). An understanding of the changes in adaptive capacity of forest systems over the course of stand development is critical to informing these strategies. Such examinations may be important given that the ability of forests to respond or adapt to rapidly changing global conditions may be intrinsic to initial stand characteristics over long time periods (Hanlon, Otto and Aitken, 2019; D'Amato and Palik, 2021).

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The uncertainties of the effects of climate change on forests have led to numerous interpretations of adaptation for natural and managed systems, but primary among these contend that high levels of compositional diversity and/or structural complexity may improve the capacity of an ecosystem to cope or respond to new or shifting conditions (Millar, Stephens and Stephenson, 2007; Evans and Perschel, 2009; Puettmann, Coates and Messier, 2009; Messier et al., 2019). These interpretations are largely based on Holling's (1973) principles of ecological resilience, which refer to the ability of an ecosystem to persist, respond, or absorb change from a disturbance without significant changes in structure and/or ecosystem function. Given the variability and uncertainty of conditions that forests may experience under global change, the concept of ecological resilience contends that higher levels of diversity may promote greater adaptive capacity to respond to change via multiple, robust pathways for recovery from disturbance or change (Holling, 1973; Carpenter et al., 2001; Elmquist et al., 2003; Gunderson, Allen and Holling, 2010). Therefore, the capacity of a forest to adaptively respond to increased variability of future climate conditions and disturbance regimes may be predicated on the diversity of life-history strategies associated with forest composition and structure to maintain core ecosystem functions in the face of these changing conditions.

The linkages between adaptive capacity and ecosystem complexity has led to an increased focus on quantifying the functional conditions characterizing a given forest ecosystem. Despite their long history of being used to describe tree species characteristics (Toumey, 1928; Violle et al., 2007; Aubin et al., 2016), plant functional traits have become increasingly applied to understand how communities of species may "respond" to environmental change or "effect" ecosystem processes (Suding et al., 2008; Curzon et al., 2017; Boisvert-Marsh et al., 2020). Effect traits include morphological (e.g., maximum height, wood specific gravity), physiological (e.g., seed mass, leaf nitrogen), and phenological (e.g., bud break, leaf longevity) attributes that characterize tree or community-level influences on an ecosystem (Lavorel and Garnier, 2002). Alternatively, response traits include a tree or community's functional tolerance or response to a given environmental filter (e.g., shade, drought, waterlogging; Violle et al., 2007). Due to their inherent link to ecological function, functional traits may be used to characterize forest development and can be integrated into species distribution models and vulnerability assessments to generate measures of future climate suitability, adaptability, or migration potential (Iverson et al., 2011; Boisvert-Marsh et al., 2020; Peters et al., 2020). In addition to trait-level analysis, measures of functional diversity (e.g., richness, dispersion, divergence, evenness; cf. Villegger, Mason and Mouillot, 2008; Laliberté and Legendre, 2010) represent useful predictive indices of ecosystem response to change since variability in overall trait diversity may confer options for adaptation to future climate conditions.

Foresters and ecologists have long used tree and stand-level structural attributes to describe aspects of forest development and productivity (McElhinny et al., 2005). The structural complexity of vertical and horizontal forest components can influence diversity in wildlife habitat, forest regeneration, hydrologic processes, nutrient cycling, and many other ecosystem functions, which independently and collectively may promote ecosystem resilience to future change (Franklin et al., 2002; Hoover and Stout, 2007; Puettmann, Coates and Messier, 2009). Stand structure may be classified in a number of ways, including but not limited to the arrangement and variability in tree sizes (e.g., diameters, heights), spatial arrangement and gap sizes, or allocation of biomass across live and/or dead trees (McElhinny et al., 2005). Measures of complexity that quantify variation in tree sizes are particularly useful as indicators of habitat heterogeneity at multiple scales, from tree (e.g., bark roughness) to stand-level (e.g., canopy stratification and niche space occupancy; Michel and Winter, 2009; Jucker, Bouriaud and Coomes, 2015). The complexity and heterogeneity of tree sizes combined with compositional diversity may increase adaptation potential in stands by increasing the variation in species, age classes, and life-history

strategies present onsite that foster additional pathways for the system to respond or recover to new climatic conditions or disturbance regimes, while simultaneously maintaining core ecosystem functions (Gunderson, 2000; Elmquist et al., 2003; Gunderson, Allen and Holling, 2010). Moreover, these aspects of forest structure and composition may also affect climate mitigation potential given their influence on patterns of aboveground biomass production and storage (Malmshiemer et al., 2008; Evans and Perschel, 2009; Fahey et al., 2010; Domke et al., 2020); however, conditions conferring maximum mitigation benefit may not always correspond with high adaptation potential (Bradford and D'Amato, 2012). As such, additional examinations are needed as to how forest developmental dynamics influence aspects of mitigation and adaptation simultaneously to inform strategies for balancing these emerging objectives (D'Amato et al., 2011).

The legacy of land use has played an important role in shaping the development of many contemporary forested landscapes (Foster, 1995; Foley et al., 2005; Ducey, Gunn and Whitman, 2013; Thompson et al., 2013). Despite the importance of climate and other biophysical drivers on influencing the future development of forest ecosystems, their capacity to respond to future change may be conditioned by the ecological inertia associated with long-term successional dynamics inherent to current forest communities developing following historic, intensive land use (Nowacki and Abrams, 2015; Pederson et al., 2015; Danneyrolles et al., 2019). Even under elevated levels of climate change, ecological simulations of future forest conditions have demonstrated that successional trajectories associated with the legacy of land use may over-ride climate impacts in the near-term (<100–200 years; Duveneck and Thompson, 2019; Nevins, D'Amato and Foster, 2021). These long-term developmental dynamics may play an important role in influencing future adaptation potential given past land use often simplifies landscape-level forest conditions (Thompson et al., 2013) both in terms of structure and composition. Moreover, large-scale anthropogenic disturbance often preferentially favors the dominance of certain taxa or species assemblages resulting in the preponderance of novel community structures that are often not maintained as systems recover over time (Hibbs, 1982; Abrams, 2001; Danneyrolles et al., 2019). Therefore, the strength of successional dynamics in second-growth forests may have long-term impacts on adaptive capacity and potential management approaches for conferring resilience in the context of global change. These factors may be uniquely important for mixed species forests that arose from past land use (e.g., *Pinus strobus* L. (eastern white pine) – *Quercus* spp. L. (oak) mixtures in New England; Kelty, 1996) or in transition ecotone forests that contain elements of both southern and northern forest communities given the documented challenges with sustaining such mixtures over time (Westveld, 1956; Hibbs, 1982; Foster and D'Amato, 2015; Kenefic et al., 2021).

The goal of this study was to examine long-term forest developmental dynamics from second-growth hardwood and softwood-hardwood mixtures of *P. strobus* – *Quercus* spp. (hereafter, referred to as mixedwood) that developed after intensive land-use disturbance in the context of adaptation and mitigation. We capitalize on a 69-year field study originally designed to evaluate facets of forest growth and yield to examine changes in forest development that influence adaptation and mitigation potential between forests and over time. The long-term nature of this field experiment afforded the opportunity to empirically evaluate changes in forests successional processes related to adaptation and mitigation. Our specific objectives were to 1) quantify the differences in stand compositional and functional diversity, structural complexity, aboveground biomass storage, and future climate suitability between forest types over time, and 2) identify the conditions that may enhance or reduce the adaptation and mitigation potential of forested communities tested. The outcomes from this work are an examination of the potential stability and/or vulnerability of second-growth transition ecotone hardwood and *P. strobus* – *Quercus* mixedwood forest types over time and the consequences for management strategies aimed at climate change adaptation and mitigation for these

communities.

2. Methods

2.1. Study area

Research was conducted in the University of Vermont Jericho Research Forest (44.445, -73.003), located 50 km west of Montpelier, Vermont, USA in the Champlain Hills Biophysical Ecoregion (Fig. 1; Thompson, Sorenson and Zaino, 2019). The site consists of second-growth transition ecotone hardwoods-white pine (*P. strobus*)-eastern hemlock (*Tsuga canadensis* (L.) Carrière) forests naturally regenerated following agricultural abandonment (Westveld, 1956). Stands that contain both hardwoods and softwoods where neither component comprises >70–80% of the composition were classified as mixedwoods for this work, after Helms (1998). Much of the research forest is situated around a hilltop plateau rising from 100 to 250 m with non-calcareous and acidic bedrock with little to no nutrient enrichment to the soil. Soils are sandy and well-drained with pH ranging from 4.5 to 7.5 with locally enriched pockets (Forrer, 2005). For the 1981–2010 period, mean temperature averaged -8.2 and 20.4 °C in January and July, respectively, while mean precipitation averaged 1039.9 mm/year⁻¹ (PRISM Climate Group).

2.2. Experimental design

In 1948, the “Farm Woodlot” experiment was established within a 16.8 ha compartment in the research forest. Upon its establishment, the experimental area is subdivided into five stands representative of common forest types found in the region. Three of the forest types include hardwood-dominated communities: two mature hardwood stands classified as 1) *mixed hardwoods* and 2) *beech-oak*, and a young, regenerating hardwood stand post-harvest referred to as 3) *cut over* (Table 1). The remaining two forest types were mixedwood stands classified as 4) *pine-hardwoods* and 5) *old field pine*. A total of thirteen 0.12–0.4 ha permanent plots were established (7 hardwood plots and 6 mixedwood plots). Based on crossdated increment cores collected from trees exhibiting characteristics of advanced age ($n = 88$, Speer, 2010), stand ages during the experiment establishment were approximately 80 years old, with the exception of the old field pine type (40 years old) and the cut over type (11 years old), the latter of which was harvested in 1937 as a 2.7 ha patch cut with *Q. rubra* and *P. strobus* residuals. An initial forest inventory was conducted in 1948 in all stands except the cut over type, which was established in 1958. All trees over 10 cm at diameter breast height (DBH) were measured, identified to species, and numerically coded with a painted ID. Inventories were conducted every five-years until 1963, when measurements were abandoned. Plots were then

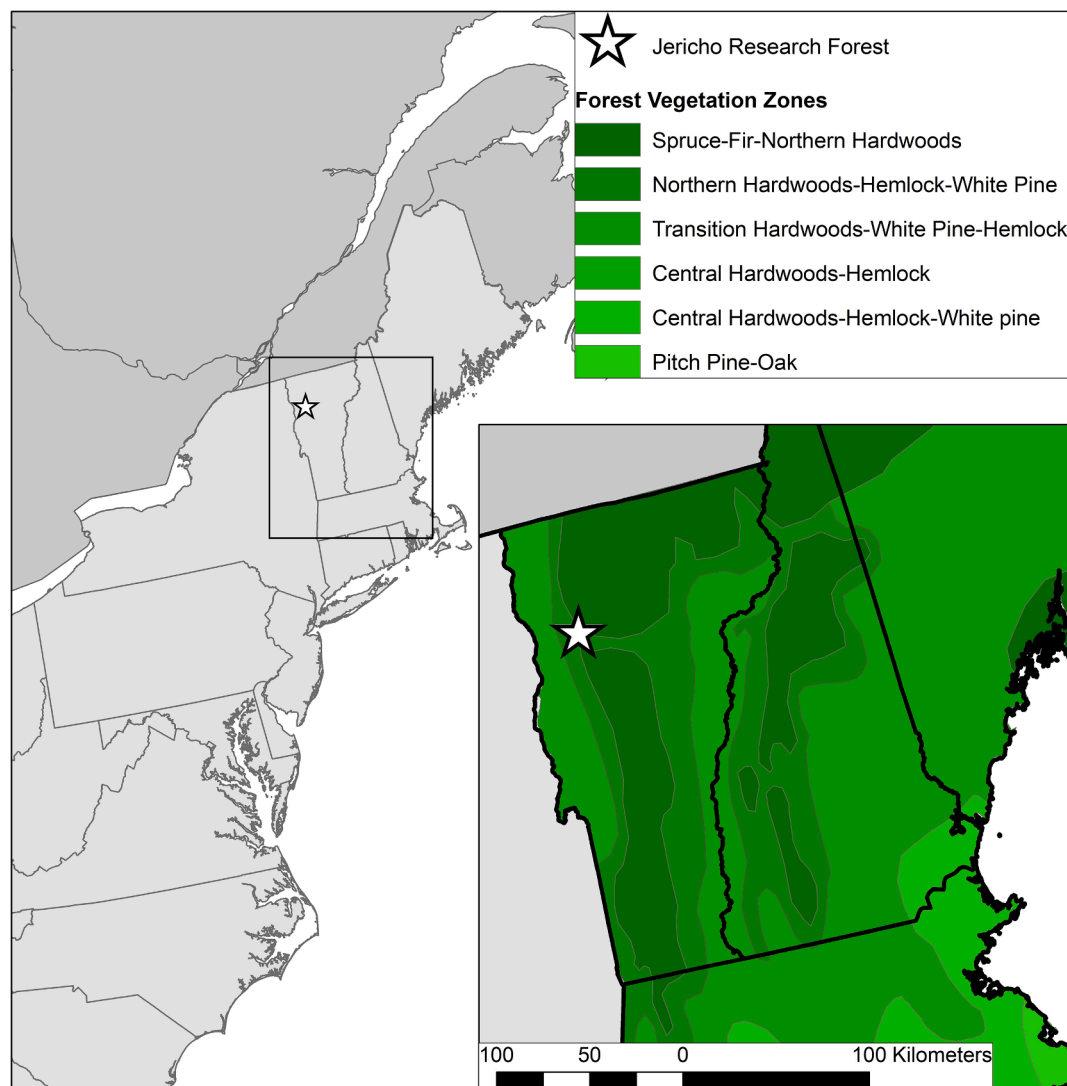


Fig. 1. Map of the Northeastern US and bordering Canadian provinces and the study site at the University of Vermont Jericho Research Forest, Vermont, US. The map inset shows regional forest vegetation zones, originally described by Westveld (1956) and reproduced by Hall and Foster (2021) for the New England states.

Table 1

Study site characteristics during the establishment of the experiment (1948–1963), including relative densities (RD), the range of harvest intensities observed among plots (d/D), overstory species importance values, and stand ages during the first census based on increment cores collected in 2017.

Compositional classification	Forest type	Plot sizes (ha)	RD (%)	d/D ^a	Species importance 1948 (%)	Approximate stand overstory age in 1948
Hardwood (n = 7)	Cut over ⁺ (n = 2)	0.20	20–39	n/a	<i>Betula</i> spp. (36.1), <i>Pinus strobus</i> (29.3), other (34.6)	11
	Beech-oak (n = 3)	0.12–0.16	40–71	0.79	<i>Fagus grandifolia</i> (27.4), <i>Quercus</i> spp. (26.2), <i>Acer</i> spp. (19.1), <i>Betula</i> spp. (10.1), other (16.8)	80
Mixedwood (n = 6)	Mixed hardwoods (n = 2)	0.20	43–69	0.32–0.73	<i>Acer</i> spp. (39.7), <i>Quercus</i> spp. (31.4), <i>Fagus grandifolia</i> (15.2), other (13.6)	80
	Old field pine (n = 2)	0.20–0.40	38–49	0.27	<i>Pinus</i> spp. (71.8), <i>Acer</i> spp. (24.6), other (3.6)	40
	Pine-hardwoods (n = 2)	0.40	26–52	0.77–1.22	<i>Pinus strobus</i> (58.1), <i>Quercus</i> spp. (17.2), <i>Acer</i> spp. (16.1), other (8.6)	80

⁺Cut over sites first censused in 1958. ^aRatio of average diameter of trees removed in harvests (d) relative to average diameter of stand prior to harvest (D). If one value is presented, only one harvest occurred during the measurement period. "Other" species = *Amelanchier* spp., *Carpinus caroliniana*, *Fraxinus americana*, *Juglans cinerea*, *Ostrya virginiana*, *Picea glauca*, *P. rubens*, *Populus* spp., *Prunus pensylvanica*, *P. serotina*, *Tilia americana*.

reestablished and remeasured in 2017, approximately 69-years after the initial establishment of the experiment. Given the minor changes in forest development during the initial fifteen-years of measurements, this study focuses on changes between the first (1948 for all forest types, except for the cut over which was established in 1958) and last (2017) measurements.

During the 2017 census, best efforts were made to reconstruct individual tree IDs for repeated measures; however, many painted IDs were indiscernible due to inherent growth of trees and spreading of the bark. Still, the persistence of paint on bark offered the opportunity to distinguish ingrowth stems that were not present during the initial inventories. Additionally, six years prior to remeasurements, a small selective harvest occurred in the pine-hardwood plots that targeted twenty large diameter trees, predominantly *P. strobus* (65%, 50.2 ± 5.9 cm DBH) and *Q. rubra* (25%, 51.3 ± 4.5 cm DBH). For these plots, stump measurements were used to reconstruct the last available DBH by species following methods described by Wharton (1984) to incorporate these data into the full 69-year analysis.

2.3. Management history

Early forest management activities in experimental stands were minor and focused on improving production capacity and promoting high-quality future sawtimber (Turner and Gibson, 1988; Forrer, 2005), primarily through the growth of *Q. rubra* and *P. strobus*. Stand improvement activities were focused on smaller size classes, as the d/D ratio was generally <1, where *d* = mean DBH of cut trees and *D* = mean stand DBH before thinning. Between 1948 and 1963, a total of only 25.8 m²/ha basal area was removed across all research plots during which time the relative density (Reineke, 1933; Woodall, Miles and Vissage, 2005) varied between forest types from 20 to 71% (Long, 1985). Records indicate that forest management activities ceased by 1984.

2.4. Structural complexity and compositional diversity

To understand changes in forest development, we chose to examine several measures of structure and composition relative to adaption potential. Although other factors such as coarse woody material and other non-tree ecosystem components play an important role in informing interpretations of forest structure and composition (Harmon et al., 1986), due to the focus of the initial study design we opted to limit this investigation to live stems over 10 cm DBH.

For our study, we define structural complexity as the stand-level heterogeneity in tree sizes based on available measurements of DBH (D'Amato et al., 2011). We also incorporate measures of density as a proxy for structure. Two measures of tree size complexity were quantified using the Gini coefficient and Shannon's diversity index. Gini's coefficient is a measure used to describe the degree of equity in a system

(Gini, 1921), and calculated as:

$$Gini = \frac{\sum_i^n (2i - n - 1)x_i}{n^2\mu}$$

where *n* is the number of trees in the plot, *x* is the basal area of tree *i*, and *μ* is the mean tree basal area. The values for the Gini coefficient range from 0 to 1, where 0 corresponds to stands with perfect size equality and 1 indicates the maximum size inequality (Weiner and Solbrig, 1984). Additionally, Shannon's diversity index (*H'*; Staudhammer and LeMay, 2001) was applied to characterize diversity in tree diameter size classes, and calculated as:

$$H_{DBH} = \sum_i^S \frac{n_i}{N} \ln \left(\frac{n_i}{N} \right)$$

where *S* is the total number of 5 cm DBH size classes in a plot, *N* is the number of trees in a plot, and *n_i* is the number of trees in size class *i*, where increasing values generally correspond to a higher number of size classes in a stand, suggesting increased structural complexity.

To examine density, we calculated relative density index based on Reineke's stand density index (Reineke, 1933; Woodall, Miles and Vissage, 2005). Live aboveground biomass estimates were also calculated for all trees using species-specific allometric equations and diameter measurements based on methods described in Chojnacki et al., (2014).

For the purposes of this study, compositional diversity refers to the abundance and diversity of tree species and plant functional groups in a plot. To examine compositional diversity, Shannon's *H'* and Simpson's *D* diversity indices were calculated for each plot in terms of species biomass using the "Vegan" package in the statistical package R version 3.6.1 (Oksanen et al., 2017; Team, 2019). Additionally, to examine the functional identity of stands, nine plant functional traits were selected for analysis based on their "effect" on or "response" to environmental factors. Mean species-specific traits were obtained from the literature, namely from a database populated by Paquette and Messier (2011; Tables A1a and A1b). Although there are numerous factors that influence forest ecosystem function that warrant exploration, the selection of traits used here was based on knowledge of the attributes that influence forest development and tree growth in this region, given the variability of species found among stands (Vielle et al., 2007). Six effect traits were tested, which include average maximum height (meters), growth rate (1: slow; 2: moderate; 3: rapid), wood density (specific gravity in g/cm³), vegetative reproduction (0: never; 1: possible; 2: common), seed mass (mg), and leaf longevity (months). Three response traits were tested and include shade, drought, and waterlogging tolerance (1: intolerant to 5: tolerant). To standardize between trait types and measurement units, z-scores were applied to a species-trait matrix to equalize the weight of traits and to meet assumptions for statistical analyses (Villegier, Mason and Mouillot, 2008). To examine the relationship of selected functional

traits to stands, community weighted means (CWM) were calculated for each plot based on mean trait values, defined as:

$$CWM_{aj} = \sum_{i=1}^n (S_{ij} * t_i)$$

where CWM_{aj} is the weighted mean for trait a for plot j , S_{ij} is the relative abundance in terms of biomass of species i in plot j , and t_i is the trait value for the species i (Lavorel et al., 2008).

Using the aforementioned functional traits, we also assessed the community-level functional diversity using four indices including functional richness (FRic), functional evenness (FEve), functional divergence (FDiv), and functional dispersion (FDis; Villegger, Mason and Mouillot, 2008; Laliberté and Legendre, 2010). FRic is analogous to species richness and refers to the volume of space occupied by functional traits present in a community when plotted in multi-dimensional space. Higher FRic is generally indicate greater variability and heterogeneity in functional traits present in a community. FEve describes the relative evenness of the distribution of traits based on their abundance distribution in a multidimensional functional trait space and is constrained from 0 (uneven distribution of traits) – 1 (even distribution of traits). FEve may become uneven when trait abundance is not consistently distributed among species. FDiv measures the divergence of extreme traits values relative to common trait values in a community, and is constrained from 0 to 1, where lower FDiv indicates higher abundance-weighted expression of common trait values, versus higher values which indicate the expression of rarer, infrequent trait values. Lastly, FDis combines the community's weighted-abundance in volume of multidimensional trait space with the spread of traits within that space and is not numerically constrained. All functional diversity indices were calculated using the R package "FD" (Laliberté, Legendre and Shipley, 2014).

To examine future climate suitability, Compatibility and Adaptability scores were calculated following methods modified from Kabrick et al., (2017). Data for both scores were obtained from the USDA Climate Change Tree Atlas Version 4 projections for the state of Vermont (Iverson et al., 2008; Peters et al., 2020). The Climate Change Tree Atlas is a species distribution model that forecasts potential changes in tree species habitat based on the current abundance by location under various greenhouse gas emission scenarios. The Tree Atlas generates species-specific Compatibility scores which are ratios of modelled future (2070–2099) suitable habitat relative to actual (2001–2016) habitat based on the sum of species importance values. Compatibility is used to indicate potential changes in species' abundance under different climate emission scenarios, where Peters et al., (2020) define habitat Compatibility classes for common species (species that occupied > 10% of modeled area) as: <0.5 = large decrease, 0.5–0.8 = small decrease, 0.8–1.2 = no change, 1.2–2.0 = small increase, >2.0 = large increase. Future habitat is modeled using an average of three general circulation models (see Peters et al., (2020) for model descriptions) under two global carbon emission scenarios, or representative concentration pathways (RCP), RCP 4.5 (low emissions) and RCP 8.5 (high emissions). The Tree Atlas also generates species-specific Adaptability scores that integrate plant functional traits and disturbance-related characteristics (e.g., susceptibility or tolerance to disease, drought, fire, pests, and pathogens) to indicate how a species may respond to future environmental change (Matthews et al., 2011). Peters et al., (2020) define Adaptability scores as 0.1–3.3 = low, 3.4–5.1 = medium, and 5.2–9.0 = high. Using both Compatibility and Adaptability scores, we calculated plot-level scores using community weighted means based on species biomass.

2.5. Statistical analysis

To test for differences in structure, composition, and climate suitability among forests tested, Analysis of Variance (ANOVA) was used

where fixed effects included broad compositional group (hardwood ($n = 7$) versus mixedwood ($n = 6$) forests) and time step (1948–2017), while forest type (cut over, beech-oak, mixed hardwood, pine-hardwood, and old field pine forest types) was included as a random effect. For some analysis, we excluded sites in young hardwood stands (cut over type) to focus our investigation on characteristics of mature hardwood forests (beech-oak, mixed hardwood; $n = 5$). All analysis were performed in the statistical software R, using the lme (Bates et al., 2015) and lmerTest (Kuznetsova, Brockhoff and Christensen, 2017) packages. A significance threshold was set for all tests at $\alpha = 0.05$ and each statistical test was assessed and diagnosed to pass test assumptions including those of linearity and normality of residuals.

3. Results

Hardwood stands (mixed hardwood, beech-oak, and cut over forest types) were initially dominated by *Betula* spp. L., *Acer* spp. L., *F. grandifolia* Ehrh., and *Quercus* spp., in small (<30 cm DBH) and intermediate (30–60 cm DBH) size classes (Fig. 2, Fig. A1). Over time, these species shifted into intermediate size classes while smaller size classes became dominated by *F. grandifolia*, *T. canadensis*, and *Acer* spp., Hardwood stands had few trees in large size classes (>60 cm), but those present were consistently *Q. rubra* and *P. strobus*. In early forest stages for mixedwood stands (pine-hardwood and old field pine forest types), *Pinus* spp. dominated across sizes including large DBH classes, *Quercus* spp. was less abundant but apparent in all size classes, and *Acer* spp. contributed to smaller classes. After 69-years, *Pinus* spp. and occasionally *Quercus* spp. dominated intermediate and larger size classes, whereas *T. canadensis*, *F. grandifolia*, *Acer* spp., and to a lesser extent, *Quercus* spp. became more numerous in the smaller and intermediate classes.

Relative to hardwoods stands, mixedwoods presently contained 37.0% more biomass on average ($\Delta 94.99 \text{ Mg ha}^{-1}$; $F = 5.73$, $p = 0.0035$; Table 2, Table A2). Although all sites exhibited increases in biomass with time ($F = 39.37$, $p < 0.001$), annual rates of accrual were lower in the mature hardwood stands (mixed hardwood and beech-oak; $1.6 \pm 0.4 \text{ Mg ha}^{-1} \text{ year}^{-1}$) compared to mixedwood stands ($3.0 \pm 0.3 \text{ Mg ha}^{-1} \text{ year}^{-1}$; $F = 6.32$, $p = 0.0033$). In the first census, biomass was predominantly in smaller DBH classes for all sites given young stand ages, but after 69-years of growth, in 2017 biomass in mixedwood sites was heavily concentrated in larger size classes (>60 cm), whereas in hardwoods it was predominately held in intermediate sizes (30–60 cm).

Irrespective of stand age, differences in Gini coefficient describing DBH distributions were apparent with greater size inequity in mixedwood stands compared to hardwood stands ($F = 6.68$, $p = 0.016$). Shannon's DBH diversity increased for all forest types through time, although mixedwood sites demonstrated the greatest complexity in size classes regardless of stand age ($F = 4.66$, $p = 0.041$).

Based on overstory tree composition, hardwood stands had higher Shannon's H' ($F = 39.29$, $p < 0.001$) and Simpson's D diversity ($F = 32.80$, $p < 0.001$) compared to mixedwood stands. In 1948, functional richness was lower in mixedwood compared to hardwood stands ($F = 4.93$, $p = 0.048$), but after 69-years this pattern was no longer apparent. All forests generally increased in functional richness, but only mixedwood sites exhibited significant increases with time (Fig. 3). Functional dispersion was consistently greater in hardwood stands compared to mixedwood stands ($F = 26.42$, $p < 0.001$). Over time, mixedwood stands became less functionally dispersed, while hardwood stands became more dispersed; however, this pattern was only significant for the former. All forests became more functionally divergent between the first measurement and the last measurement, although this was only significant in mixedwoods. No differences were observed in functional evenness.

Based on community weighted means, mixedwood stands had greater values for species effect traits such as maximum heights ($F = 42.83$, $p < 0.001$), growth rates ($F = 39.22$, $p < 0.001$), and leaf

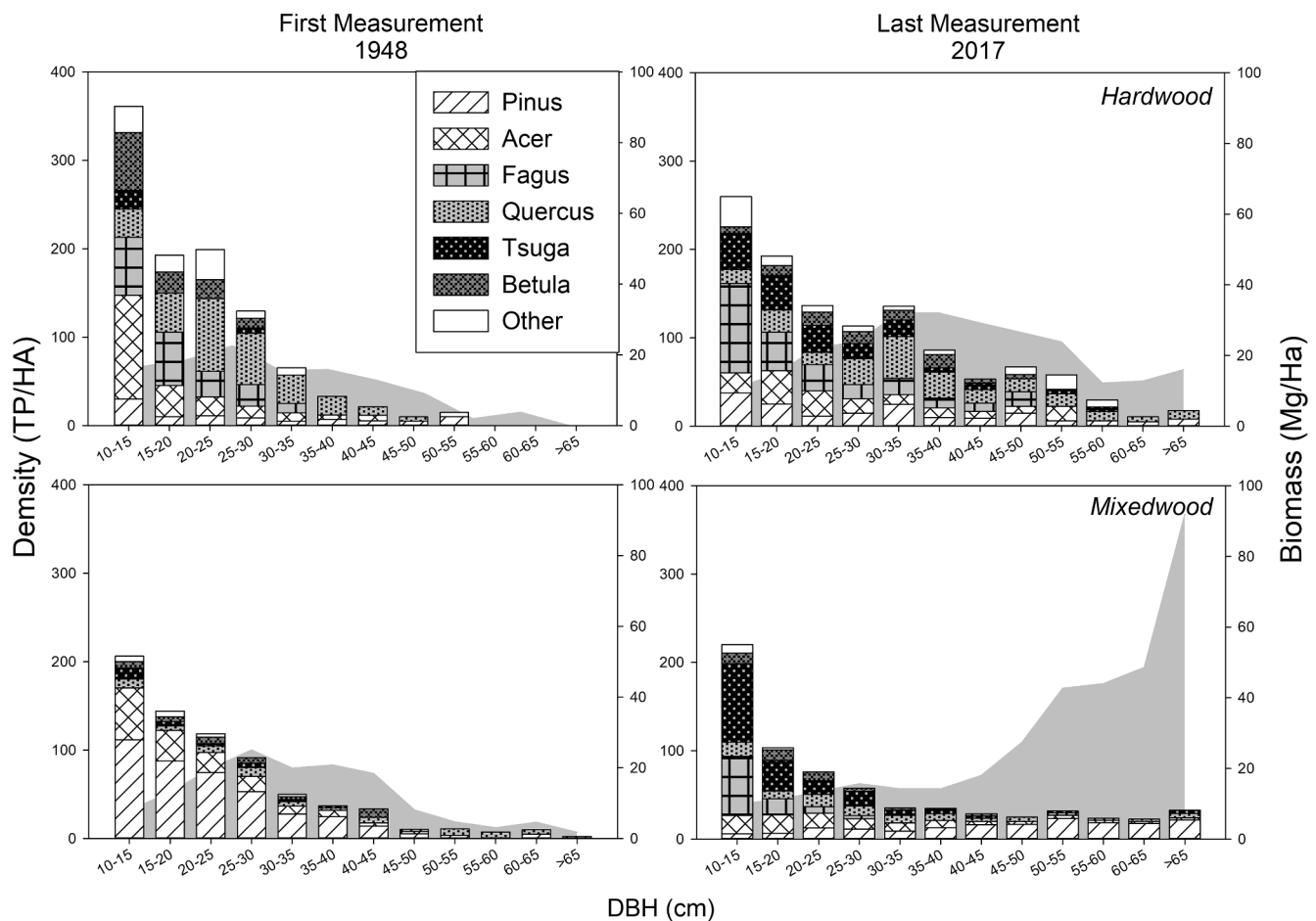


Fig. 2. Diameter distributions and live aboveground biomass across 5 cm diameter classes (DBH) between hardwood and mixedwood forests tested (horizontal panels) and two measurement periods (1948–2017; vertical panels). Shaded gray area corresponds to biomass. See Fig. A1 for diameter distributions by individual forest type. Species grouped by genera: *Pinus* = *P. strobus* and *P. resinosa*; *Acer* = *A. rubrum*, *A. saccharum*, and *A. pensylvanicum*; *Fagus* = *F. grandifolia*; *Quercus* = *Q. rubra* and *Q. alba*; *Tsuga* = *T. canadensis*; *Betula* = *B. papyrifera*, *B. lenta*, and *B. alleghaniensis*; Other = *Amelanchier* spp., *Carpinus caroliniana*, *Fraxinus americana*, *Juglans cinera*, *Ostrya virginiana*, *Picea glauca*, *P. rubens*, *Populus* spp., *Prunus pensylvanica*, *P. serotina*, *Tilia americana*.

longevity ($F = 58.86$, $p < 0.001$) compared to hardwood stands, which were dominated by traits such as greater wood densities ($F = 61.39$, $p < 0.001$), seed masses ($F = 12.15$, $p = 0.001$), and the capacity to vegetatively resprout ($F = 48.69$, $p < 0.001$). The difference in response traits was less pronounced among forest communities, although hardwood communities were consistently more waterlogging tolerant compared to mixedwoods ($F = 5.26$, $p < 0.031$). Mature hardwoods (mixed hardwoods and beech-oak) were more shade tolerant than mixedwoods ($F = 7.46$, $p < 0.013$). No differences were observed in drought tolerance between forest types.

The functional profiles of overstory trees in all forests were relatively stable over time (Fig. 4). No changes in the overstory composition of hardwoods were observed, but mixedwood stands increased in shade tolerance, leaf longevity, and height, while decreased in waterlogging tolerance. Ingrowth over the 69-year measurement period resulted in substantial changes for nearly all traits for mixedwoods while hardwoods were again relatively unchanged. Ingrowth trees in mixedwood stands were increasingly shade and waterlogging tolerant with higher wood densities, seed masses, and capacities to vegetatively resprout as well as decreasing drought tolerance and lower growth rates. Conversely, ingrowth in hardwood stands only decreased in waterlogging tolerance.

Future climate Adaptability and Compatibility varied by forest type and time. Overall, most stands exhibited moderate levels of Adaptability (range 3.4–5.1; Fig. 5; Matthews et al., 2011; Peters et al., 2020), except for early forest stages of the mixed hardwoods type which had high

Adaptability scores (>5.1). Irrespective of measurement period, hardwood stands had higher Adaptability than mixedwoods ($F = 10.66$, $p = 0.003$). In early forest stages, Adaptability scores for hardwoods varied more than mixedwoods, but differences in scores stabilized over time. Both mature hardwood (beech-oak, mixed hardwood forest types) and mixedwood forests showed reductions in Adaptability with time ($F = 7.83$, $p = 0.011$), although this trend was most pronounced in mixedwood stands. Compatibility scores followed similar trends as Adaptability, as most of the forest types tested were forecasted to experience positive increases in future habitat suitability across the emissions scenarios we tested. No differences in Compatibility were detected between forest types across stand ages. Under present stand conditions, hardwood sites were expected to have greater future climate Compatibility than mixedwoods under both the low (RCP4.5: Δ Compatibility = 1.13; $F = 5.36$, $p = 0.041$) and high greenhouse gas emission scenarios (RCP8.5: Δ Compatibility = 1.47; $F = 6.04$, $p = 0.032$). When climate suitability scores of ingrowth are compared against those of established trees on site, ingrowth in hardwood stands have lower Compatibility and Adaptability relative to established stems, but there was no apparent difference within mixedwood stands (See Fig. A1).

4. Discussion

Studies of forest structural and compositional change over time have long been used to understand patterns of forest development and guide management (Fisher and Terry, 1920; Oliver and Stephens, 1977) with

Table 2

Structural, compositional, and functional characteristics and climate suitability scores for hardwood and mixedwood forest communities tested at the initial (1948) and final (2017) measurement period. All values are arithmetic means and $\pm$ standard errors. See Table A2 for table by individual forest type.

Forest Development	Hardwood		Mixedwood	
	1948*	2017	1948	2017
Structure				
Gini Index	0.40 $\pm$ 0.01	0.50 $\pm$ 0.01	0.45 $\pm$ 0.01	0.60 $\pm$ 0.01
Shannon's H_{DBH}	1.28 $\pm$ 0.15	1.92 $\pm$ 0.07	1.78 $\pm$ 0.02	2.09 $\pm$ 0.06
Biomass (Mg/ha)	116.63 $\pm$ 21.30	256.80 $\pm$ 24.59	147.56 $\pm$ 10.20	351.79 $\pm$ 27.03
Relative Density (%)	47.39 $\pm$ 6.42	63.42 $\pm$ 4.23	42.41 $\pm$ 2.28	54.23 $\pm$ 2.54
Composition				
Functional Richness	16.64 $\pm$ 2.95	22.35 $\pm$ 2.50	6.66 $\pm$ 2.84	25.50 $\pm$ 1.42
Functional Evenness	0.56 $\pm$ 0.03	0.44 $\pm$ 0.07	0.49 $\pm$ 0.08	0.47 $\pm$ 0.06
Functional Divergence	0.81 $\pm$ 0.03	0.87 $\pm$ 0.02	0.85 $\pm$ 0.05	0.96 $\pm$ 0.01
Functional Traits (CWMs)				
Growth Rate	0.16 $\pm$ 0.15	-0.03 $\pm$ 0.12	0.90 $\pm$ 0.08	0.77 $\pm$ 0.01
Wood Density	0.14 $\pm$ 0.16	0.18 $\pm$ 0.10	-0.71 $\pm$ 0.11	-0.87 $\pm$ 0.04
Max Height	0.39 $\pm$ 0.05	0.42 $\pm$ 0.04	0.63 $\pm$ 0.03	0.72 $\pm$ 0.02
Veg. Reproduction	0.54 $\pm$ 0.21	0.55 $\pm$ 0.20	-0.55 $\pm$ 0.09	-0.77 $\pm$ 0.04
Seed Mass	0.57 $\pm$ 0.22	0.65 $\pm$ 0.14	0.08 $\pm$ 0.13	0.04 $\pm$ 0.03
Leaf Longevity	-0.23 $\pm$ 0.10	-0.11 $\pm$ 0.10	0.45 $\pm$ 0.07	0.71 $\pm$ 0.04
Shade Tolerance	-0.02 $\pm$ 0.13	0.22 $\pm$ 0.15	-0.13 $\pm$ 0.02	0.03 $\pm$ 0.02
Drought Tolerance	-0.37 $\pm$ 0.18	-0.52 $\pm$ 0.18	-0.37 $\pm$ 0.07	-0.41 $\pm$ 0.03
Waterlogging Tolerance	-0.10 $\pm$ 0.19	-0.02 $\pm$ 0.15	-0.23 $\pm$ 0.08	-0.58 $\pm$ 0.05
Climate Suitability (CWMs)				
Compatibility RCP 4.5	2.75 $\pm$ 0.68	2.67 $\pm$ 0.41	1.66 $\pm$ 0.24	1.54 $\pm$ 0.09
Compatibility RCP 8.5	3.03 $\pm$ 0.87	2.89 $\pm$ 0.50	1.52 $\pm$ 0.25	1.42 $\pm$ 0.12
Adaptability	5.00 $\pm$ 0.24	4.84 $\pm$ 0.09	4.68 $\pm$ 0.13	3.82 $\pm$ 0.10

Community weighted mean = CWM. Functional trait CWM values are presented as z-scores to standardize between trait types and measurement units.

* All forests were sampled in 1948, except for the cut over forest type which was first sampled in 1957.

recent attention on the relationship between developmental patterns and a forest's capacity to adapt to or mitigate the effects of future anticipated climate change (Millar, Stephens and Stephenson, 2007; Evans and Perschel, 2009). Given the importance of historic land use in affecting current forest conditions in many regions of the globe, an understanding of the linkages between secondary forest development and levels of adaptive capacity are critical to informing vulnerability assessments and developing adaptation strategies. Although much attention has been paid to the development of second-growth forests in northeastern North America and other regions with respect to demography, structure, and productivity, key knowledge gaps remain regarding how these developmental dynamics influence future climate adaptation or mitigation potential (D'Amato et al., 2011). This study focused on changes within a relatively small area; however, its long-term, empirical nature allowed for a demonstration of the influence of forest composition and stand age on future climate suitability, adaptability, and mitigation of the forest types tested. Specifically, our results show that the developmental patterns of hardwood stands tested are

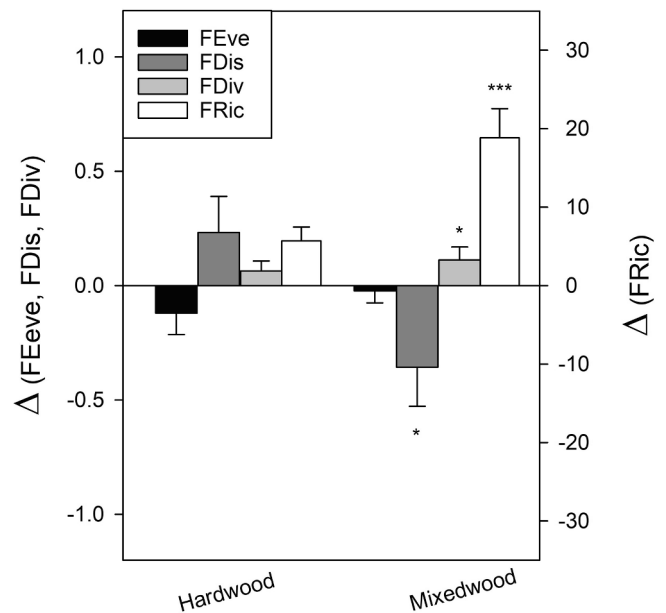


Fig. 3. Absolute change (Δ) in functional diversity over 69-years for hardwood and mixedwood forest types. Error bars are standard errors and asterisks indicate a change in index value between time steps. Note change in y-axis between diversity indices. $p \leq 0.05 = *$, $p \leq 0.01 = **$, $p \leq 0.001 = ***$ Abbreviations for functional diversity indices: FEve = evenness, FDis = dispersion, FDiv = diversion, and FRic = richness.

more compositionally stable over time, relative to mixedwood communities, which are more divergent and transitory with important implications for the long-term maintenance and adaptation trajectory of these forests types.

Maturation of forest communities has generally been linked with a degree of stability in compositional and functional diversity, particularly as shade-tolerant species become more dominant in the absence of large-scale disturbance (Bu, Zang and Ding, 2014; Frelich, 2016; Vicente-Silva et al., 2016; Curzon et al., 2017). The forests examined in this work showed some propensity for increases in compositional diversity with stand ages up to 80–150 years, with the overall increase in functional richness across all forests over time corresponding to later successional conditions (Leak, 1987; Bormann and Likens, 1994; Oliver and Larson, 1996; Frelich, 2016; Franklin, Johnson and Johnson, 2018). Of the forest systems examined, the greatest gains in functional richness were in mixedwood stands likely due to lower levels of richness during stand initiation (primarily the old field pine type) coupled with substantial increases in structural complexity. Trends in functional dispersion over time were positive for hardwoods and negative for mixedwoods with the divergence in these trends related to an increase of like traits in hardwoods compared to a reduced weighting of traits that typify mixedwoods, namely the conifer components. The changes in functional diversity observed in the mixedwood stands tested here correspond with observations elsewhere that highlight the transitory nature of mixedwoods forest types arising from anthropogenic disturbance, with these systems frequently trending towards a greater preponderance of hardwood species and traits over time (Hibbs, 1982; Kenefic et al., 2021; Kern et al., 2021). While *T. canadensis* increased the softwood component in these sites, the shifts in functional traits associated with ingrowth on mixedwood sites not only trended towards higher shade tolerance, but largely favored those traits of hardwood species such as increased wood densities, seed masses, and a capacity to vegetatively resprout. It is likely that without intervention to promote the regeneration and maintenance of characteristic *P. strobus-Quercus* mixedwood species, the long-term developmental trajectory of these stands will continue to trend towards that more closely resembling the

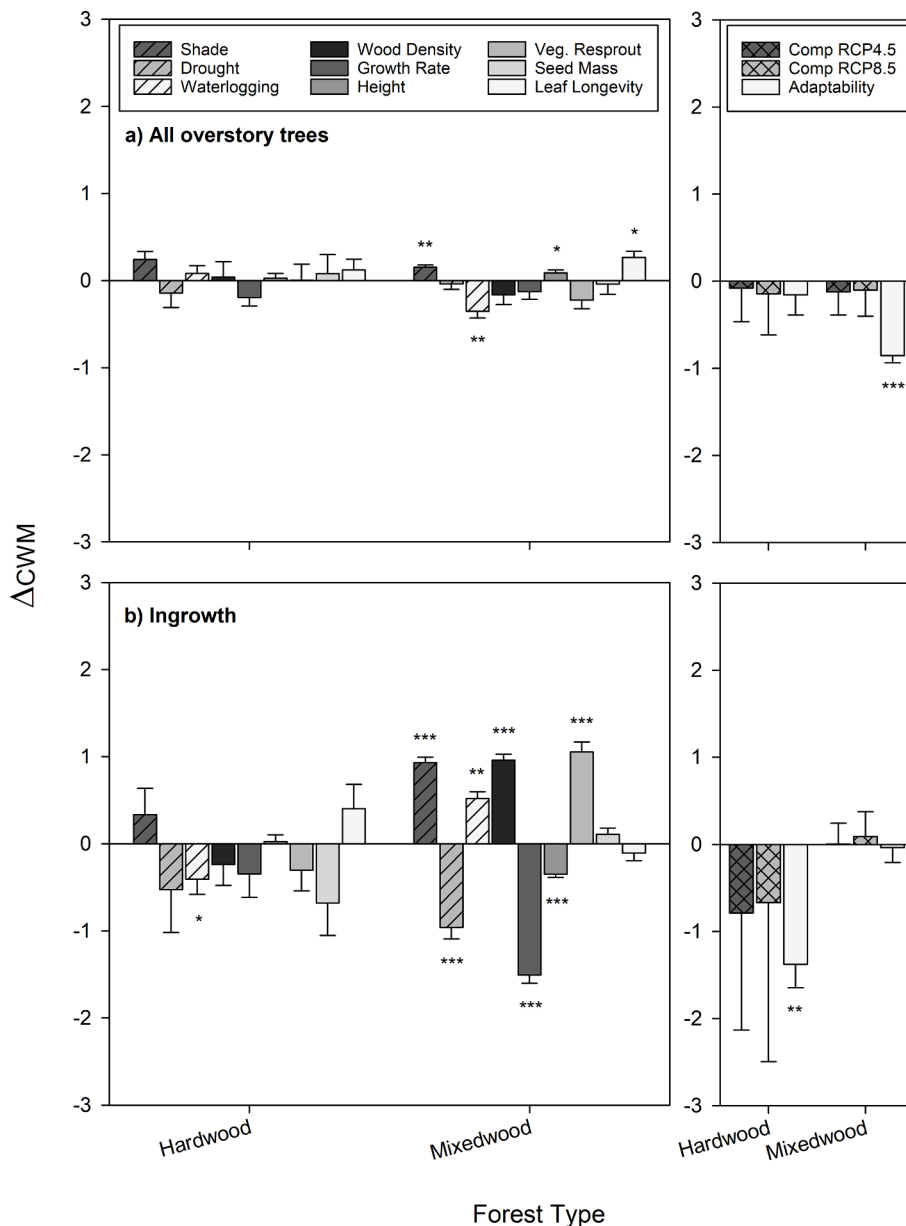


Fig. 4. Change (Δ) in plant functional traits (left column) and future climate (2070–2099) suitability scores (Compatibility and Adaptability; right column) based on community weighted means (CWM) in terms of species biomass over time for hardwood (cut over, beech-oak, mixed hardwood) and mixedwood (old field pine, pine-hardwood) forest types. a) Summarizes changes for all trees in stands, and b) represents the differences between ingrowth trees (≥ 10 cm DBH established after 1948) relative to trees present in 2017 census that established prior to 1948. Compatibility scores are presented under a low emission scenario (RCP 4.5) and high emission scenario (RCP 8.5). Error bars are standard errors and asterisks indicate Δ in index value between 1948 and 2017 measurements. $p \leq 0.05 = *$, $p \leq 0.01 = **$, $p \leq 0.001 = ***$

traits of hardwood dominated stands (cf. Kenefic et al., 2021).

The changes in composition and functional identity over time strongly influence the adaptive capacity and climate suitability of the systems we examined. In particular, *F. grandidentata* and *T. canadensis*, two of the most shade-tolerant species in the region, comprised 64% of all ingrowth trees and led to increases in overall shade tolerance across all sites, a pattern characteristic of later successional stands in this region (Cline and Spurr, 1942; Ducey, Gunn and Whitman, 2013; Waterman et al., 2020). Compared to species frequently in the overstory in these stands (*A. rubrum*, *Q. rubra*, and *P. strobus*), *F. grandidentata* and *T. canadensis* are expected to decline in future habitat suitability (mean Compatibility = 0.66, or “small decrease”) and have biological and disturbance characteristics that are only moderately adaptive (mean Adaptability = 3.9) relative to the dominant canopy species on these sites (pooled Compatibility = 2.4, or “large increase”; pooled Adaptability = 5.7, or “high adaptability”). Furthermore, *F. grandidentata* is affected by beech bark disease (*Cryptococcus fagisuga* Lind. and *Neonectria* spp.) and *T. canadensis* is affected by hemlock woolly adelgid (*Adelges tsugae* Annand), non-native invasive insects and pathogens that

have negatively impacted the survival and growth of these species throughout much of their range (Shigo, 1972; Orwig and Foster, 1998; Lovett et al., 2016). As such, these shifts in functional profiles and species composition towards that of increased shade tolerance may not only lead to reduced future climate suitability, but also a greater vulnerability to the impacts of non-native insects and diseases.

Forest biomass (or carbon) accretion and storage are recognized as important climate mitigation strategies to offset greenhouse gas emissions (Evans and Perschel, 2009; Fahey et al., 2010). Our results report similar levels of total live aboveground biomass storage in second-growth hardwood and mixedwood stands as similarly-aged stands in the region (Fahey, Sherman and Weinstein, 2013; Eisen and Plotkin, 2015; Kabrick et al., 2017; Urbano and Keeton, 2017; Finzi et al., 2020). Compared to biomass stores in the hardwood stands, mixedwoods consistently held nearly 100 Mg ha⁻¹ more biomass. This pattern was most apparent in the pine-hardwood mixedwood forest types examined in this work, which compared to the mature hardwood communities tested supported approximately 54% more aboveground live biomass (183 $\pm$ 20.8 Mg ha⁻¹), and even approached levels found in some old-

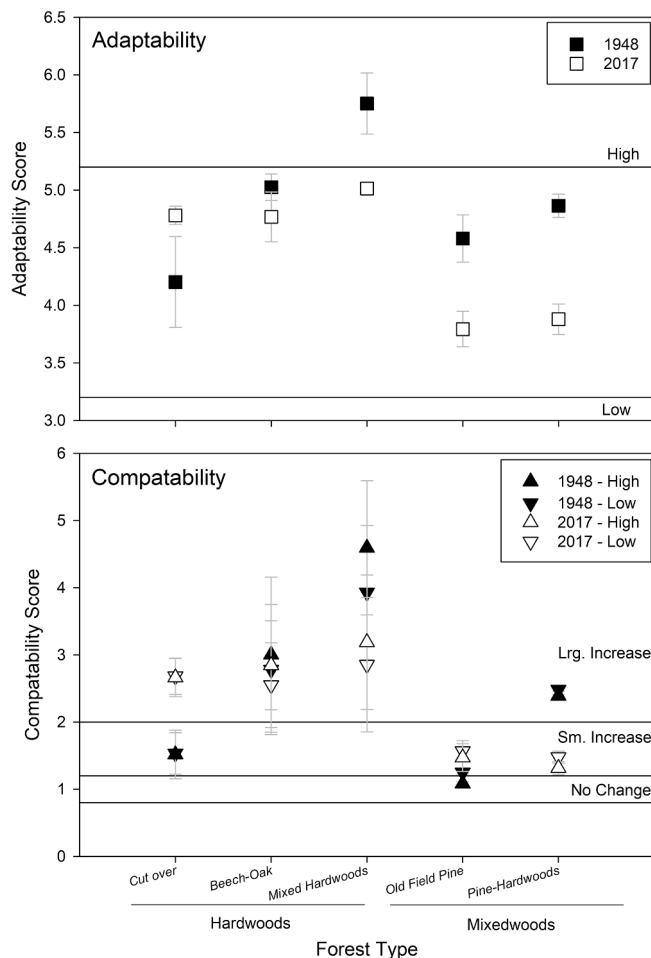


Fig. 5. Future climate Adaptability and Compatibility scores for five hardwood and mixedwood forest types examined at the initial (1948) and final (2017) measurement. Compatibility scores are species-specific ratios of modelled future (2070–2099) suitable habitat relative to actual (2001–2016) habitat used to indicate potential changes in each species' abundance under two global carbon emission scenarios: RCP 4.5 (low emission scenario) or RCP 8.5 (high emission scenario). Adaptability scores are species-specific values that integrate plant functional traits and disturbance-related characteristics to indicate how a species may respond to future environmental change (Kabrick et al., 2017). Scores are presented based on community weighted means in terms of species biomass.

growth stands in the region (D'Amato et al., 2017; Waterman et al., 2020). This pattern can primarily be attributed to the presence of *P. strobus* in the upper canopy layers (68% of biomass in stands), a finding that corroborates the importance of this species in contributing toward large biomass stores in stratified, mixed species stands (see further discussion below; Seymour, 2011; D'Amato et al., 2017; Waterman et al., 2020).

Our findings support the well documented positive relationship between aboveground biomass and stand age (Pregitzer and Euskirchen, 2004), although clear differences in both biomass accretion rate and overall storage among hardwood and mixedwood forests are likely attributed to differences in growth strategies, variability in structural complexity, and the presence/absence of larger trees found in these communities. For example, mixedwood forests, such as the *Pinus-Quercus* systems in this study, are generally more structurally complex and vertically stratified given softwood and hardwood trees may more effectively occupy space due to differing life-history traits, like shade-tolerance, growth rate, and crown structure (Hibbs, 1982; Kelty, Larson, B.C. and Oliver, 1992; Waskiewicz et al., 2013; Kabrick et al., 2017;

Kenefic et al., 2021). The capacity for overyielding in *P. strobus*-*Q. rubra* mixtures has been observed elsewhere (Waskiewicz et al., 2013), likely due to the differences in crown architecture and the canopy positions occupied by these two species even in single-cohort stands. This combined with the increased structural complexity and propensity for larger diameter trees in these stands, particularly tall and fast-growing *P. strobus* and to a lesser extent *Q. rubra* with broad canopy form and high wood density, led to nearly twice as much biomass sequestered and stored in mixedwood stands tested, particularly the pine-hardwood forest types.

The importance of structural complexity and large diameter trees in affecting aboveground biomass stores correspond with findings from second-growth and old-growth stands in the region that demonstrate the disproportionate role of large trees, specifically upper-canopy *P. strobus* in generating exceedingly large biomass values relative to those stands where large *P. strobus* was absent (Lutz et al., 2012; Eisen and Plotkin, 2015; D'Amato et al., 2017; Waterman et al., 2020). While *T. canadensis* provides an additional softwood component in the mid-canopy in the stands tested, the potential contribution of this species in terms of biomass is quite low relative to that of large *P. strobus*, which has been shown elsewhere (Kabrick et al., 2017; Waterman et al., 2020). These findings are important in the context of management for climate mitigation aimed at increasing forest carbon, since stratified mixed-species forests that contain large diameter canopy *P. strobus* may provide the greatest potential for maximizing aboveground carbon sequestration and stores. Paradoxically, when vast stores (here, over 65%) are concentrated in just a few, large trees from one species, the risk of catastrophic losses due to stochastic disturbance-related mortality, emerging forest health issues (Wyka et al., 2017), or selective timber harvests targeting the largest trees highlights the high vulnerability of these stand attributes (Foster, 1988; Kenefic et al., 2021). This is particularly salient given the potential increased frequency of natural disturbances, such as wind and insects, which may target large trees or a single species reducing the role of these structural components in enhancing carbon stocks (Dale et al., 2001; Bradford et al., 2013; D'Amato et al., 2017; Santoro and D'Amato, 2019). Furthermore, given that many species comprising subordinate canopy positions are less future climate suitable and susceptible to issues of forest health, the compounding effects of these disturbances could reduce both the forest mitigation and adaption potential.

4.1. Conclusions and management implications

The hardwood and mixedwood stands examined here exhibit differing and complementary qualities important for future climate adaptation and mitigation potential, but the pattern of long-term compositional stability of hardwood stands compared to the transitory nature of *P. strobus*-*Quercus* mixedwoods highlight the likely need for management strategies to maintain these stands to sustain ecological and climate benefits. Despite the limited scope of this study, the relative stability of the hardwood sites examined coupled with high future climate suitability suggest these transition ecotone hardwood forest types may persist, maintain, and possibly increase in abundance into the future. These results corroborate findings from other hardwood forests that show strong ecological memory (Johnstone et al., 2016; Webster et al., 2018) and resilience mechanisms to disturbance (e.g., capacity for vegetative resprouting, advance regeneration; Plotkin et al., 2013; Santoro and D'Amato, 2019) that confer high adaptive capacity of these communities. Conversely, the successional dynamics in the *P. strobus*-*Quercus* mixedwoods, species mixtures that were likely a minor component in the region prior to European settlement and are primarily an artifact of intensive land use (Kelty, 1996; Abrams, 2001), make these novel communities much more ephemeral over longer time periods (Kenefic et al., 2021). While the mixedwood stands exhibited increased compositional and functional diversity, structural complexity, and biomass sequestered and stored, we observed little to no regeneration of

P. strobus and *Quercus* spp. over the 69-years, pointing to a potential decrease in the abundance of these communities over time.

Our findings examining long term *P. strobus-Quercus* mixedwood development indicate that the maintenance of these forest communities may require deliberate silvicultural intervention, such as site preparation of seedbed conditions and release treatments over time, to sustain *Pinus strobus* and *Quercus* spp., especially on rich sites where other hardwood species may outcompete them (Cline and Lockard, 1925; Hibbs, 1982; Kenefic et al., 2021). To achieve multiple benefits, these strategies may be coupled with silvicultural frameworks that adaptively plan and respond to future conditions (Joyce et al., 2009; Kabrick et al., 2017; Nagel et al., 2017; Palik et al., 2020). Although tradeoffs have been shown to exist when attempting to achieve multiple conflicting objectives (D'Amato et al., 2011; Bradford and D'Amato, 2012), compared to hardwood-dominated communities, the *P. strobus-Quercus* mixedwoods tested here appear to balance high functional diversity and biomass storage, bridging an important gap for managers seeking to maintain high adaptation potential with climate mitigation.

CRedit authorship contribution statement

Peter W. Clark: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft. **Anthony W. D'Amato:** Conceptualization, Methodology, Resources, Writing – review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Available from ScienceBase.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foreco.2021.119654>.

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